

Cisco 4000 Family Integrated Services Router

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Cisco® 4000 Family Integrated Services Routers (ISRs) form a Software Defined WAN platform that delivers the performance, security, and convergence capabilities that today's branch offices need.

Product overview

The Cisco 4000 Family Integrated Services Router (ISR) revolutionizes WAN communications in the enterprise branch. With new levels of built-in intelligent network capabilities and convergence, it specifically addresses the growing need for application-aware networking in distributed enterprise sites. These locations tend to have lean IT resources. But they often also have a growing need for direct communication with both private data centers and public clouds across diverse links, including Multiprotocol Label Switching (MPLS), VPNs and the Internet.

The Cisco 4000 Family Integrated Services Router (ISR) contains the following platforms:

ISR4461, 4451, 4431, 4351, 4331, 4331-DC, 4321, 4221 & 4221X.



Figure 1.
Cisco 4000 Series Integrated Services Routers

Features and benefits

Cisco 4000 Family ISRs provide you with Cisco® Catalyst Software Defined WAN (SD-WAN) software features and a converged branch infrastructure. Along with superior throughput, these capabilities form the building blocks of next-generation branch-office WAN solutions.

Cisco Catalyst Software Defined WAN

Cisco Catalyst SD-WAN is a set of intelligent software services that allow you to reliably and securely connect users, devices, and branch office locations across a diverse set of WAN transport links. Cisco Catalyst SD-WAN enabled routers like the ISR 4000 dynamically route traffic across the “best” link based on up-to-the-minute application and network conditions for great application experiences. You get tight control over application performance, bandwidth usage, data privacy, and availability of your WAN links - control that you need as your branches conduct greater volumes of mission-critical business.

Cisco converged branch infrastructure

The Cisco 4000 Series ISRs consolidate many must-have IT functions, including network, compute, and storage resources. The high-performance, integrated routers run multiple concurrent services, including encryption, traffic management, and WAN optimization, without slowing your data throughput. And you can activate new services on demand through a simple licensing change.

Cisco intent based networking and Digital Network Architecture (Cisco DNA)

The last few years has seen a rapid transformation and adoption of digital technologies. This puts pressure on the on the network teams supporting this changing infrastructure – especially when provisioning, managing, monitoring and troubleshooting these diverse devices. Additionally, innovations such as Cisco Catalyst SD-WAN, Network Function Virtualization (NFV), Open APIs and Cloud Management show great promise in transforming IT networks. This transformation raises further questions and challenges for the IT teams.

The Cisco Digital Network Architecture (Cisco DNA) is an open, extensible, software-driven architecture that provides for faster innovation, helping to generate deeper insights, and deliver exceptional experiences across many different applications. Cisco DNA relies on intent-based networking, a revolutionary approach in networking that helps organizations automate, simplify, and secure the network.

The intent-based Cisco DNA network is:

- **Informed by Context:** Interprets every byte of data that flows across it, resulting in better security, more customized experiences, and faster operations.
- **Powered by Intent:** Translates your intent into the right network configuration, making it possible to manage and provision multiple devices and things in minutes.
- **Driven by Intuition:** Continually learns from the massive amounts of data flowing through it and turns that data into actionable insight. Helps you solve issues before they become problems and learn from every incident.

Cisco Catalyst Center provides a centralized management dashboard across your entire network – the branch, campus, data center, and cloud. Rather than relying on box-by-box management, you can design, provision, and set policy end-to-end from the single Cisco Catalyst Center interface. This allows you to respond to organizational needs faster and to simplify day-to-day operations. Cisco Catalyst SD-WAN Analytics, Assurance and Cisco Network Data Platform (NDP) help you get the most from your network by continuously collecting and putting insights into action. Cisco DNA is open, extensible, and programmable at every layer. It integrates Cisco and third-party technology, open APIs, and a developer platform, to support a rich ecosystem of network-enabled applications.

Table 1 breaks out many of the features and benefits of the Cisco 4000 Family that create a Catalyst SD-WAN and a converged branch infrastructure.

Table 1. Cisco 4000 Family ISR general feature highlights

Business Requirement(s)	Feature/Solution
Performance • Throughput • Service reliability	• Concurrent software services at speeds up to 10 Gbps (ISR4461). Backplane architecture supports high-bandwidth module-to-module communication at speeds up to 10 Gbps. • A distributed multicore architecture with the industry's first internal services plane. • Remote installation of application-aware services, which run identically to their counterparts in dedicated appliances.
Lower WAN expenditures	• Embedded Cisco Catalyst SD-WAN solution for creating lower-cost, business-class Internet connections.

Business Requirement(s)	Feature/Solution
Pay-as-you-grow • Performance upgrade model • Investment protection • CapEx budget management	• Router capacity can be increased with a remote performance-on-demand license upgrade (no hardware upgrade) for exceptional savings.
Superior and secure user application experiences	• ISR-AX “Application Experience” software bundle with advanced routing and network monitoring services. • Dynamic Multipoint VPN (DMVPN), zone-based firewalls, Intrusion Prevention (Snort and Umbrella Branch) and Content Management using Cisco Cloud Web security and OpenDNS protecting data, providing authentication credentials, and transmissions not backhauled through the data center. • Secure boot feature performs hardware-based authentication of the bootloader software to prevent malicious or unintended software from booting on the system. • Code signing verifies digital signatures of executables prior to loading to prevent execution of altered or corrupted code. • Hardware authentication protects against hardware counterfeiting by using an on-board tamper-proof silicon, including field replaceable modules. If authentication fails, the module is not allowed to boot.
IT consolidation, space savings, and improved Total Cost of Ownership (TCO)	• Single converged branch platform integrates routing, switching, virtual server, storage, security, unified communications, WAN optimization, and performance management tools.
Business continuity and increased resiliency	• ISR 4400 Series models (4461, 4451, and 4431 ISRs) support dual integrated power supplies for backup. The entire ISR 4000 Family supports optional power supply capable of delivering additional PoE power to endpoints. • Modular network interfaces with diverse connection options for load-balancing and network resiliency. • Modular interfaces with online removal and insertion (OIR) for module upgrades without network disruption. • Cisco Unified Survivable Remote Site Telephony (SRST), which serves as a resiliency complement to Cisco Hosted Collaboration Solution (HCS), a Cisco cloud-based UC service. • Support for multiple, diverse access links: T1/E1, T3/E3, Serial, xDSL, Gigabit and Ten-Gigabit Ethernet.
Lower telephony costs with VoIP and rich media experiences	• High-performance analog/digital gateway, allowing VoIP over less expensive Session Initiation Protocol (SIP) trunks. • Integrated IP PBX (Cisco Unified Communications Express) and Session Border Controller (Cisco Unified Border Element, or CUBE).
Easier manageability and support	• Single, universal software image for all features and performance-on-demand licensing flexibility. • No additional services and support needed for compute and storage. • Supported by Cisco and third-party management tools, with programmability and automation.

Platform architecture

Table 2 lists the primary hardware architectural features and benefits of the Cisco 4000 Family. The routers run modular Cisco IOS XE Software, widely deployed in the world's most demanding networks. The software's comprehensive portfolio of services spans multiple technology areas, including security, WAN optimization, app and network Quality of Service (QoS), and embedded management.

Table 2. Architectural highlights

Architectural Features	Benefits/Description
Multicore processors	• High-performance multicore processors support high-speed WAN connections. The data plane uses an emulated Flow Processor (FP) that delivers Application-Specific Integrated Circuit (ASIC)-like performance that does not degrade as services are added.
Embedded IP Security (IPsec) VPN hardware acceleration	• Increases scalability. When combined with an optional Cisco IOS XE Software Security license, enables WAN link security and VPN services.
Integrated Gigabit Ethernet ports	• The Cisco 4000 Family provides up to six built-in Ethernet ports for WAN or LAN. • Depending on platform, Ethernet ports can support Small Form-Factor Pluggable (SFP)-based connectivity as well as RJ-45 connections, enabling fiber or copper connectivity. • Optionally, depending on the platform, up to 30W PoE+ can be enabled on two of the built-in front panel Gigabit Ethernet interfaces to provide power to external devices such as 4G/LTE/5G Cellular Gateways. • An additional dedicated Gigabit Ethernet port is provided for device management¹.
USB-based console access	• A mini type-B USB console port¹ supports management connectivity when traditional serial ports are not available. • Traditional console and auxiliary ports are also available².
Optional integrated power supply for distribution of PoE	• An optional upgrade to the internal power supply provides inline power (802.3af-compliant PoE or 802.3at-compliant PoE+) to optional integrated switch modules. • Redundant PoE conversion modules provide an additional layer of fault tolerance.
Optional integrated Redundant Power Supply (RPS)	• For the ISR 4400 Series, power redundancy is available by installing an optional power supply, protecting the network from power failures. • Optional PoE boost mode increases total PoE capacity to up to 1000W.
Cisco Enhanced Services Module (SM-X)	• Each service-module slot offers high data-throughput capability of up to 10 Gbps toward the system and up to 1 Gbps to other module slots. • Support for both single- and double-wide service modules provides flexibility in deployment options. • An SM-X slot can be converted into a Network Interface Module (NIM) slot using an optional carrier card. • Service modules support Online Insertion and Removal (OIR), avoiding network disruption when installing new or replacement modules¹.

¹ Not supported on ISR4221

² ISR4221 supports shared Console & Auxiliary ports

Architectural Features	Benefits/Description
Cisco Network Interface Modules (NIMs)³	- Up to three integrated NIM slots on the Cisco 4000 Family allow for flexible configurations. - Each NIM slot offers options of up to two 2Gbps connections. One towards the route processor and one for direct module to module communication. The ISR 4221 has only one 1Gbps connection to the Route Processor. - NIMs support OIR. - Special NIMs add support Solid-State Drives (SSDs) and Hard Disk Drives (HDDs)¹.
Cisco Integrated Services Card (ISC) slot on motherboard	- Integrated Services Card natively supports the new Cisco High-Density Packet Voice Digital Signal Processor Modules (PVD4s), providing greater-density rich-media voice. - Each Integrated Services Card slot connects to the system architecture through an up-to 2-Gbps link. - Future modules can be hosted on the Integrated Services Card slot, improving system functions.
Flash memory support	- A single flash memory slot is available to support high-speed storage densities, upgradable to up to 32 GB. The ISR4221 ships with a fixed quantity of 8GB of flash memory. - Two USB type A 2.0 ports provide capabilities for convenient storage¹.
DRAM	- For the ISR 4400 Series, the default control-plane memory is 4 GB, upgradable to 16 GB to provide additional scalability for control-plane features. Data-plane memory is 2 GB, non-upgradable. - For the ISR 4300 Series, the default memory is 4 GB, upgradable to 16 GB (only 8GB for 4321) to provide additional scalability. - The ISR 4200 Series comes in two versions: ISR4221 - 4GB fixed DRAM and ISR4221X - 8GB fixed DRAM.
NEBS (Network Equipment Building System)	The ISR 4461 is NEBS certified to maintain reliable operations in harsh and demanding environments. A NEBS kit is orderable with dual 650W-DC reverse-airflow power supplies and a reverse airflow fan assembly.

Managing your Cisco ISR 4000 Family ISRs

The Cisco network management applications listed at the top of Table 3 are standalone products that can be purchased or downloaded to manage your Cisco network devices. The applications are built specifically for the different operational phases; select those that best fit your needs. Those management capabilities listed under the “Cisco IOS Software XE Embedded Management” heading are directly integrated into the routers’ software operating system.

³ Unified Communications (UC) and UC based NIM's are not supported

Table 3. Cisco network management applications

Operational Phase	Management Task by Application	Description
Device staging and configuration	WebUI	• A GUI-based device-management tool for Cisco IOS and Cisco IOS XE Software-based access routers. This tool simplifies routing, firewall, VPN, unified communications, and WAN and LAN configuration through easy-to-use wizards.
Network-wide deployment, configuration, monitoring, and troubleshooting	Cisco Catalyst Center	• Offers comprehensive lifecycle management of wired and wireless access, campus, and branch-office networks, rich visibility into end-user connectivity, and application performance assurance. • Provides wired lifecycle functions such as inventory, configuration, and image management; automated deployment; compliance reporting; integrated best practices; and reporting.
Staging, deployment, and changes to configuration and image files	Cisco Configuration Engine	• A secure network management product that provides zero-touch image and configuration distribution through centralized, template-based management.
Context-aware security configuration and monitoring	Cisco Prime Security Manager	• Management tool for configuring and managing context-aware security. The application supports both single- and multi-device manager form factors. • Provides the ability to write and enforce the granular context-aware security policies.
Cisco Wide Area Application Service (WAAS) management	Cisco WAAS Central Manager	• The management tool for the WAAS^{1 4}, (WAN optimization and application acceleration) integrated service. It provides a centralized mechanism for configuring WAAS features, reporting, and monitoring.

Table 4. Cisco IOS XE Software Embedded Management Capabilities

Feature	Description
Cisco IOS Embedded Event Manager (EEM)	• A distributed and customized approach to event detection and recovery. • Offers the ability to monitor events and take informational, corrective, or any desired EEM action when the monitored events occur or when a threshold is reached.
Cisco IOS XE IP Service-Level Agreements (IP SLAs)	• Helps assure the performance of new business-critical IP applications as well as IP services that use data and voice in an IP network.
SNMP , Remote Monitoring (RMON), syslog , NetFlow , IP Flow Information Export (IPFix)	• Network monitoring and accounting tools.

⁴ It is suggested to use AppNav with an External WAAS device for the ISR4221

Product specifications

Table 5 lists the general product specifications for the Cisco 4000 Family routers.

Table 5. Specifications of Cisco 4000 Family integrated services routers

Technical Specifications	Cisco 4461	Cisco 4451	Cisco 4431	Cisco 4351	Cisco 4331/ 4331-DC	Cisco 4321	Cisco 4221(X)
Aggregate Throughput (Default)	1.5Gbps	1 Gbps	500 Mbps	200 Mbps	100 Mbps	50 Mbps	35Mbps
Aggregate Throughput (Performance License)	3Gbps	2 Gbps	1 Gbps	400 Mbps	300 Mbps	100 Mbps	75 Mbps
Aggregate CEF Only ⁵ Throughput (Boost License)	Over 7Gbps	Over 4Gbps	Over 4Gbps	Over 2Gbps	Over 2Gbps	1.5Gbps	1.2Gbps
Total onboard WAN or LAN 10/100/1000 ports	4	4	4	3	3	2	2
Total onboard WAN or LAN 10Gbps ports	2	-	-	-	-	-	-
RJ-45-based ports	4	4	4	3	2	2	2
SFP-based ports	4	4	4	3	2	1	1
Enhanced service-module slots	3	2	0	2	1	0	0
Doublewide service-module slots	2	1 (assumes no singlewide SM-X modules installed)	0	1 (assumes no singlewide SM-X modules installed)	0	0	0
NIM slots	3	3	3	3	2	2	2
OIR (all I/O modules)	Yes	Yes	Yes	Yes	Yes	Yes	No
Onboard ISC slot	1	1	1	1	1	1	No

⁵ Using onboard Gigabit Ethernet Interfaces

Technical Specifications	Cisco 4461	Cisco 4451	Cisco 4431	Cisco 4351	Cisco 4331/ 4331-DC	Cisco 4321	Cisco 4221(X)
Default memory double-data-rate 3 (DDR3) error-correction-code (ECC) DRAM (Combined control/services/ data planes)	NA	NA	NA	4 GB	4 GB	4 GB	4GB 8GB (4221X)
Maximum memory DDR3 ECC DRAM (Combined control/services/ data planes)	NA	NA	NA	16 GB	16 GB	8 GB	4GB 8GB (4221X)
Default memory DDR3 ECC DRAM (data plane)	4 GB	2 GB	2 GB	NA	NA	NA	NA
Maximum memory DDR3 ECC DRAM (data plane)	4 GB	2 GB	2 GB	NA	NA	NA	NA
Default memory DDR3 ECC DRAM (control/services plane)	8 GB	4 GB	4 GB	NA	NA	NA	NA
Maximum memory DDR3 ECC DRAM (control/services plane)	32 GB	16 GB	16 GB	NA	NA	NA	NA
Default flash memory	8 GB	8 GB	8 GB	4 GB	4 GB	4 GB	8GB
Maximum flash memory	32 GB	32 GB	32 GB	16 GB	16 GB	8 GB	8GB
External USB 2.0 slots (type A)	2	2	2	2	1	1	1
USB console port -type B mini (up to 115.2 kbps)	1	1	1	1	1	1	0
Serial console port - RJ45 (up to 115.2 kbps)	1	1	1	1	1	1	1 (combo CON/AUX port)
Serial auxiliary port - RJ45 (up to 115.2 kbps)	1	1	1	1	1	1	1 (combo CON/AUX port)

Technical Specifications	Cisco 4461	Cisco 4451	Cisco 4431	Cisco 4351	Cisco 4331/ 4331-DC	Cisco 4321	Cisco 4221(X)
Power-supply options	Internal: AC, DC and PoE	Internal: AC, DC and PoE	Internal: AC, DC, and PoE	Internal: AC, DC and PoE	Internal: AC and PoE, DC* (*4331-DC)	External: AC and PoE	External AC only
Redundant power supply	Internal: AC, DC and PoE	Internal: AC, DC and PoE	Internal: AC, DC, and PoE	N/A	N/A	N/A	NA
Power Specifications							
AC input voltage	100 to 240 VAC autoranging	100 to 240 VAC autoranging	100 to 240 VAC autoranging	100 to 240 VAC autoranging	100 to 240 VAC autoranging	100 to 240 VAC autoranging	100 to 240 VAC autoranging
DC Input Voltage					(4331-DC)		
Input Voltage	48 – 60V	48 – 60V	48 – 60V	NA	24 – 60V	NA	NA
Input Current	12A Max	12A Max	12A Max	NA	14 – 5A	NA	NA
AC input frequency	47 to 63 Hz	47 to 63 Hz	47 to 63 Hz	47 to 63 Hz	47 to 63 Hz	47 to 63 Hz	47 to 63 Hz
AC input current range, AC power supply (maximum)	7.1 to 3.0A	7.1 to 3.0A	3 to 1.3A	7.1 to 3.0A	3 to 1.3A	1.5 to 0.6A	1.5 to 0.6A
AC input surge current	60 A peak and less than 5 Arms per half cycle	<50 A	60 A peak and less than 5 Arms per half cycle	60 A peak and less than 12 Arms per half cycle	60 A peak and less than 5 Arms per half cycle	90 A peak and less than 3 Arms per half cycle	90 A peak and less than 3 Arms per half cycle
Typical power (no modules) (watts)		158	65	48	42	36	24
Maximum power with AC power supply (watts)	1000W (no PoE)	450 (no PoE)	250 (no PoE)	430	250	125	90
Maximum power with DC power supply (watts)	437 (no PoE)	437 (no PoE)	437 (no PoE)	437 (no PoE)	250 (no PoE) 4331-DC: PoE not supported		
Maximum power with PoE power supply (platform only) (watts)	1000 with PoE redundant 1450 with PoE boost no redundancy	1000 with PoE redundant 1450 with PoE boost no redundancy	500 with PoE redundant 1000 with PoE boost no redundancy	990	530 4331-DC: PoE not supported	260	NA (No PoE Support)
Maximum endpoint PoE power available from PoE power supply (watts)	500 W with optional redundancy	500 W with optional redundancy	250 W with optional redundancy	500	250 4331-DC: PoE not supported	120	NA (No PoE Support)

Technical Specifications	Cisco 4461	Cisco 4451	Cisco 4431	Cisco 4351	Cisco 4331/ 4331-DC	Cisco 4321	Cisco 4221(X)
Maximum endpoint PoE power capacity with PoE boost (watts)	950 W no redundancy	950 W no redundancy	500 W no redundancy	N/A	N/A	N/A	NA (No PoE Support)
Sizes and Weights							
Dimensions (H x W x D)	3.5 x 17.25 x 18.5 in (88.9 x 438.15 x 469.9 mm)	3.5 x 17.25 x 18.5 in (88.9 x 438.15 x 469.9 mm)	1.73 x 17.25 x 19.97 in (43.9 x 438.15 x 507.2 mm)	3.5 x 17.25 x 18.5 in (88.9 x 438.15 x 469.9 mm)	1.75 x 17.25 x 17.25 in (44.45 x 438.15 x 438.15 mm)	1.75 x 14.55 x 11.60 in (44.55 x 369.57 x 294.64 mm)	1.72 x 12.7 x 10" (43.7 x 322.6 x 254 mm)
External Power Supply Dimensions (H x W x D)	N/A	N/A	N/A	N/A	N/A	2.95 x 1.18 x 6.10 in (75 x 30 x 155 mm)	37 x 73 x 152 mm (Phihong mfg PN: AA90U-120A-R) 36.5 x 67 x 155 mm (Delta mfg PN: ADP90GR BA)
Shipping Box Dimensions (H x W x D)	9.75 x 22.25 x 26 in (24.76 x 56.51 x 66.04 mm)	9.75 x 22.25 x 26 in (24.76 x 56.51 x 66.04 mm)	7.88 x 22.25 x 28.75 in (200.2 x 565.1 x 730.25 mm)	9.75 x 22.25 x 26 in (24.76 x 56.51 x 66.04 mm)	7.125 x 22.75 x 22.5 in (180.98 x 577.85 x 571.5 mm)	7.0 x 21.5 x 16.125 in (177.8 x 546.1 x 409.6 mm)	4.13 x 18.25 x 12.94" (104.9 x 463.6 x 328.7 mm)
Rack height	3 Rack Units (3RU)	2 Rack Units (2RU)	1 Rack Units (1RU)	2 Rack Units (2RU)	1 Rack Unit (1RU)	1 Rack Unit (1RU)	1 Rack Unit (1RU)
Rack-mount 19in. (48.3 cm) EIA	Included	Included	Included	Included	Included	Included	Optional
Rack-mount 23in. (58.4 cm) EIA	Optional	Optional	Optional	Optional	Optional	N/A	NA
Wall-mount	No	No	Yes	No	Yes	Mounting holes under chassis	Yes
Weight with 1, 450-WAC power supply (no modules)		28.8 lb (13.1 kg)	N/A	28.8 lb (13.1 kg)	N/A	N/A	NA
Weight with 1 1,000-WAC power supply+ 1 PoE power module (no other modules)		30.6 lb (13.9 kg)	N/A	29.0 lb (13.2 kg)	N/A	N/A	NA

Technical Specifications	Cisco 4461	Cisco 4451	Cisco 4431	Cisco 4351	Cisco 4331/ 4331-DC	Cisco 4321	Cisco 4221(X)
Weight with AC PS (no modules)		N/A	18.5 lb (8.4 kg)	N/A	13.5 lb (6.2 kg)	7.7 lb (3.5 kg) + 1.2 lb (0.66 kg) external PS	7.1 lb (3.22kg)
Weight with DC PS (no modules)		28.8lb (13.1kg)	28.8lb (13.1kg)	28.8lb (13.1kg)	13.5 lb (6.2 kg) (4431-DC)		
Weight with AC PS with POE (no modules)		N/A	18.6 lb (8.4 kg)	N/A	14.1 lb (6.4 kg)	N/A	NA
Typical weight (fully loaded with modules)		42.7 lb (19.4 kg)	22.4 lb (10.2 kg)	37.7 lb (17.1 kg)	16.1 lb (7.3 kg)	9.14 lb (4.2 kg) + 1.2 lb (0.66 kg) external PS	8.11 lb (3.68kg)
Packaging Weight		6.4 lb (2.9 kg)	5.9 lb (2.7 kg)	6.4 lb (2.9 kg)	4.6 lb (2.1 kg)	2.2 lb (1 kg)	1.28 lb (0.58kg)
Airflow	I/O side to bezel side	I/O side to bezel side	I/O side to bezel side	I/O side to bezel side	I/O side to bezel side	Right I/O side to Left I/O side	I/O side to bezel side
MTBF (Hours)	480770	480770	512970	566310	587250	593270	593270
Environmental Specifications							
Operating Conditions							
Temperature	32 to 104°F (0 to 40°C)	32 to 104°F (0 to 40°C)	32 to 104°F (0 to 40°C)	32 to 104°F (0 to 40°C)	32 to 104°F (0 to 40°C)	32 to 104°F (0 to 40°C)	32 to 104°F (0 to 40°C)
Altitude (China)	0 – 6,560 ft. (0 – 2,000 m)	0 – 6,560 ft. (0 – 2,000 m)	0 – 6,560 ft. (0 – 2,000 m)	0 – 6,560 ft. (0 – 2,000 m)	0 – 6,560 ft. (0 – 2,000 m)	0 – 6,560 ft. (0 – 2,000 m)	0 – 6,560 ft. (0 – 2,000 m)
Altitude (Rest of the world)	0 – 10,000 ft. (0 – 3,050 m)	0 – 10,000 ft. (0 – 3,050 m)	0 – 10,000 ft. (0 – 3,050 m)	0 – 10,000 ft. (0 – 3,050 m)	0 – 10,000 ft. (0 – 3,050 m)	0 – 10,000 ft. (0 – 3,050 m)	0 – 10,000 ft. (0 – 3,050 m)
Short Term Temperature upto 6560ft per NEBS GR-63-CORE	NA	NA	NA	NA	32° to 131° (0° to 55° C) (for ISR4331-DC version)	NA	NA
Relative humidity	5% to 85%	5% to 85%	5% to 85%	5% to 85%	5% to 85%	5% to 85%	5% to 85%
Short-term humidity	5% to 90%, not to exceed 0.024 kg water/kg of dry air	5% to 90%, not to exceed 0.024 kg water/kg of dry air	5% to 90%, not to exceed 0.024 kg water/kg of dry air	5% to 90%, not to exceed 0.024 kg water/kg of dry air	5% to 90%, not to exceed 0.024 kg water/kg of dry air	5% to 90%, not to exceed 0.024 kg water/kg of dry air	5% to 90%, not to exceed 0.024 kg water/kg of dry air
Acoustics: Sound pressure	50.6/73.1 dBA	50.6/73.1 dBA	54.3/79.1 dBA	50.6/73.1 dBA	52.8/74.8 dBA	24.2/51.9 dBA	28.5/53 dBA

Technical Specifications	Cisco 4461	Cisco 4451	Cisco 4431	Cisco 4351	Cisco 4331/ 4331-DC	Cisco 4321	Cisco 4221(X)
(Typical/ maximum)							
Acoustics: Sound power (Typical/ maximum)	Z	58.2/78.8 dBA	57.2/80.8 dBA	58.2/78.8 dBA	61.2/81.6 dBA	31.9/59.9 dBA	41/68 dBA
Nonoperating Conditions							
Temperature	-40 to 158°F (-40 to 70°C)	-40 to 158°F (-40 to 70°C)	-40 to 158°F (-40 to 70°C)	-40 to 158°F (-40 to 70°C)	-40 to 158°F (-40 to 70°C)	-40 to 158°F (-40 to 70°C)	-40 to 158°F (-40 to 70°C)
Relative humidity	5% to 95%	5% to 95%	5% to 95%	5% to 95%	5% to 95%	5% to 95%	5% to 95%
Altitude	15,584 ft (4750m)	15,584 ft (4750m)	15,584 ft (4750m)	15,584 ft (4750m)	15,584 ft (4750m)	15,584 ft (4750m)	15,584 ft (4750m)
Regulatory and Compliance							
Safety	UL 60950-1 CAN/CSA C22.2 No. 60950-1 EN 60950-1 AS/NZS 60950-1 IEC 60950-1	UL 60950-1 CAN/CSA C22.2 No. 60950-1 EN 60950-1 AS/NZS 60950-1 IEC 60950-1	UL 60950-1 CAN/CSA C22.2 No. 60950-1 EN 60950-1 AS/NZS 60950-1 IEC 60950-1	UL 60950-1 CAN/CSA C22.2 No. 60950-1 EN 60950-1 AS/NZS 60950-1 IEC 60950-1	UL 60950-1 CAN/CSA C22.2 No. 60950-1 EN 60950-1 AS/NZS 60950-1 IEC 60950-1	UL 60950-1 CAN/CSA C22.2 No. 60950-1 EN 60950-1 AS/NZS 60950-1 IEC 60950-1	UL 60950-1 CAN/CSA C22.2 No. 60950-1 EN 60950-1 AS/NZS 60950-1 IEC 60950-1
EMC	47 CFR, Part 15 ICES-003 Class A EN55032 Class A CISPR32 Class A AS/NZS CISPR 32 Class A VCCI V-3 CNS 13438 EN 300-386 EN 61000 (Immunity) EN 55024, CISPR 24 KN22, KN24	47 CFR, Part 15 ICES-003 Class A EN55032 Class A CISPR32 Class A AS/NZS CISPR 32 Class A VCCI V-3 CNS 13438 EN 300-386 EN 61000 (Immunity) EN 55024, CISPR 24 KN22, KN24	47 CFR, Part 15 ICES-003 Class A EN55032 Class A CISPR32 Class A AS/NZS CISPR 32 Class A VCCI V-3 CNS 13438 EN 300-386 EN 61000 (Immunity) EN 55024, CISPR 24 KN22, KN24	47 CFR, Part 15 ICES-003 Class A EN55032 Class A CISPR32 Class A AS/NZS CISPR 32 Class A VCCI V-3 CNS 13438 EN 300-386 EN 61000 (Immunity) EN 55024, CISPR 24 KN22, KN24	47 CFR, Part 15 ICES-003 Class A EN55032 Class A CISPR32 Class A AS/NZS CISPR 32 Class A VCCI V-3 CNS 13438 EN 300-386 EN 61000 (Immunity) EN 55024, CISPR 24 KN22, KN24 EN 50121-1 EN61000-6-4	ICES-003 Class A EN55032 Class A CISPR32 Class A AS/NZS CISPR 32 Class A VCCI V-3 CNS 13438 EN 300-386 EN 61000 (Immunity) EN 55024, CISPR 24 KN22, KN24	ICES-003 Class A EN55032 Class A CISPR32 Class A AS/NZS CISPR 32 Class A VCCI V-3 CNS 13438 EN 300-386 EN 61000 (Immunity) EN 55024, CISPR 24 KN22, KN24

Technical Specifications	Cisco 4461	Cisco 4451	Cisco 4431	Cisco 4351	Cisco 4331/4331-DC	Cisco 4321	Cisco 4221(X)
					EN 50121-4		
Telecom	T1	T1	TIA-968-B	TIA-968-B	TIA-968-B	TIA-968-B	TIA-968-B
	IC CS-03:2004	IC CS-03:2004	CS-03	CS-03	CS-03	CS-03	CS-03
	TIA-968-B:2009	TIA-968-B:2009	ANSI T1.101	ANSI T1.101	ANSI T1.101	ANSI T1.101	ANSI T1.101
	HKTA 2028:2010	HKTA 2028:2010	ITU-T G.823, G.824	ITU-T G.823, G.824	ITU-T G.823, G.824	ITU-T G.823, G.824	ITU-T G.823, G.824
	HKTA 2017:2010	HKTA 2017:2010	IEEE 802.3	IEEE 802.3	IEEE 802.3	IEEE 802.3	IEEE 802.3
	HKTA 2015:2006	HKTA 2015:2006	RTTE Directive	RTTE Directive	RTTE Directive	RTTE Directive	RTTE Directive
	G.703:2001	G.703:2001	Homologation requirements vary by country and interface type. For specific country information, refer to the online approvals data base at: <u>https://www.cisco.com/</u> .	Homologation requirements vary by country and interface type. For specific country information, refer to the online approvals data base at: <u>https://www.cisco.com/</u> .	Homologation requirements vary by country and interface type. For specific country information, refer to the online approvals data base at: <u>https://www.cisco.com/</u> .	Homologation requirements vary by country and interface type. For specific country information, refer to the online approvals data base at: <u>https://www.cisco.com/</u> .	Homologation requirements vary by country and interface type. For specific country information, refer to the online approvals data base at: <u>https://www.cisco.com/</u> .
	ID0002:2007	ID0002:2007					
	IS6100:2004	IS6100:2004					
	DSPR Gray Book:2000	DSPR Gray Book:2000					
	DSPR Technical Condition: 2004	DSPR Technical Condition: 2004					
	E1	E1					
	AS/ACIF S016: 2001	AS/ACIF S016: 2001					
	AS/ACIF S038: 2001	AS/ACIF S038: 2001					
	G.703:2001	G.703:2001					
	TBR 4:1995	TBR 4:1995					
	TBR 12:1993	TBR 12:1993					
	TBR 13:1996	TBR 13:1996					
	RRA 2009-38	RRA 2009-38					
	(RRL 2005-96)	(RRL 2005-96)					
	IDA TS DLCN:2011	IDA TS DLCN:2011					
	IDA TS ISDN PRA:2005	IDA TS ISDN PRA:2005					
	IS6100: 2004	IS6100: 2004					
	PTC 220:2008	PTC 220:2008					
	Ethernet	Ethernet					
	IEEE 802.3	IEEE 802.3					
	ANSA	ANSA					

Technical Specifications	Cisco 4461	Cisco 4451	Cisco 4431	Cisco 4351	Cisco 4331/ 4331-DC	Cisco 4321	Cisco 4221(X)
	X3.263						
Cisco IOS XE Software							
Protocols	IPv4, IPv6, static routes, Routing Information Protocol Versions 1 and 2 (RIP and RIPv2), Open Shortest Path First (OSPF), Enhanced IGRP (EIGRP), Border Gateway Protocol (BGP), BGP Router Reflector, Intermediate System-to-Intermediate System (IS-IS), Multicast Internet Group Management Protocol Version 3 (IGMPv3), Protocol Independent Multicast sparse mode (PIM SM), PIM Source Specific Multicast (SSM), RSVP, CDP, ERSPAN, IPSLA, Call Home, EEM, IKE, ACL, EVC, DHCP, FR, DNS, LISP, OTV ⁶ , HSRP, RADIUS, AAA, AVC, Distance Vector Multicast Routing Protocol (DVMRP), IPv4-to-IPv6 Multicast, MPLS, Layer 2 and Layer 3 VPN, IP sec, Layer 2 Tunneling Protocol Version 3 (L2TPv3), Bidirectional Forwarding Detection (BFD), IEEE802.1ag, and IEEE802.3ah						
Encapsulations	Generic routing encapsulation (GRE), Ethernet, 802.1q VLAN, Point-to-Point Protocol (PPP), Multilink Point-to-Point Protocol (MLPPP), Frame Relay, Multilink Frame Relay (MLFR) (FR.15 and FR.16), High-Level Data Link Control (HDLC), Serial (RS-232, RS-449, X.21, V.35, and EIA-530), and PPP over Ethernet (PPPoE)						
Traffic management	QoS, Class-Based Weighted Fair Queuing (CBWFQ), Weighted Random Early Detection (WRED), Hierarchical QoS, Policy-Based Routing (PBR) and NBAR.						
Cryptographic Algorithms	Encryption: DES, 3DES, AES-128 or AES-256 (in CBC and GCM modes); Authentication: RSA (768/1024/2048 bit), ECDSA (256/384 bit); Integrity: MD5, SHA, SHA-256, SHA-384, SHA-512						

Services plane: Enabling the Branch-in-a-Box

All Cisco ISR 4000 routers contain processing cores built-in as standard to allow full-featured services to run on-board. This includes the full-featured Cisco WAAS engine that provides application acceleration and highly responsive virtual desktop experience. The technology is known as Cisco Service Containers and it uses a standard hypervisor to allow x64 based applications to run.

The 4000 series routers can be fitted with Solid State Drives (SSD) and server cards for local storage and computing capability. The Cisco UCS-E server cards are available with 8-core Intel Xeon processors with up to 48GB of high speed DDR3 memory and three drives built in offering RAID 0, 1 and 5. This immense amount of compute power can eliminate the need for any dedicated servers at branch sites. UCS-E cards can be configured and managed using VMware vCenter and pooled with Data Center compute resources.

⁶ Supported only on the 4451 for Bandwidths <100Mbps

Software Subscription through Cisco DNA Licensing

The ISR4000 series supports software-based subscription using Cisco DNA based licensing. Two Cisco DNA software subscription licenses are available for the WAN portfolio: Cisco DNA Essentials and Cisco DNA Advantage.

Cisco DNA SD-WAN and Routing licenses

There are two Cisco DNA SD-WAN and Routing licenses available:

- Cisco DNA Essentials (Subscription) + Network Essentials (Perpetual)
- Cisco DNA Advantage (Subscription) + Network Advantage (Perpetual)

Cisco DNA Essentials offers:

- A maximum of 10 User VPNs + 1 Management VPN
- Cisco Catalyst SD-WAN Manager for centralized management (cloud or on-premises)
- Flexible topology (hub and spoke, partial mesh, full mesh)
- Application-based policies (including application-aware routing policies)
- Essential Cisco Catalyst SD-WAN security services including
 - L3/L4/App-Aware Firewall
 - Snort IPS/IDS with Talos signature updates
 - URL-filtering
 - Cisco Advanced Malware Protection (AMP)
 - Cisco Umbrella cloud-app discovery
 - SD-WAN Application Intelligence Engine (SAIE)
- Essential Cloud Networking
 - **Multicloud:** GCP, AWS, Azure
 - **SaaS:** All applications
- DNS monitoring and connector for Cisco Umbrella
- Basic path optimization capabilities including Forward Error Correction (FEC)
- Dynamic routing protocols (OSPF/BGP)

Cisco DNA Advantage adds:

- Unlimited segmentation
- Cisco Catalyst SD-WAN Analytics
- Predictive Path Recommendations (PPR), powered by ThousandEyes WAN Insights
- Advanced Cloud OnRamp for Multicloud and SaaS (all applications and telemetry)
- Automated Service Stitching

For more information on Cisco DNA for SD-WAN and Routing subscriptions, please refer to <https://www.cisco.com/site/us/en/products/networking/wan-software/dna-subscription-wan/index.html>.

For a more detailed overview on Cisco Licensing, go to cisco.com/go/licensingguide.

Enterprise NFV on ISR4000

Built to reduce costs without compromising vital network services, the UCS E-Series router-integrated branch blade servers provide support for a Virtualization-ready and Application-centric platform that can be seamlessly integrated on the ISR4000 platform. Customers can now install virtualized applications on the ISR4000 routers through the Cisco® Enterprise NFV Infrastructure Software (NFVIS) – a virtualization infrastructure that integrates full VM lifecycle management, monitoring, device programmability, and service chaining in a single, installable package. For more information on Enterprise NFV and NFVIS, please refer to the link at <https://www.cisco.com/c/en/us/solutions/enterprise-networks/enterprise-network-functions-virtualization-nfv/index.html#~stickynav=2>.

Support for DC Power

ISR4000 platforms support both DC and AC power supplies as options. Specifically, the ISR 4331 has two separate product SKU's – the ISR4331/K9 and the ISR4331-DC/K9 which support AC and DC power respectively. The ISR4400 can independently support an AC or a DC power supply on the same chassis. While the 4300 supports between 24V and 60Vdc, the 4400 supports between 48V and 60Vdc. The 4331 provides for up to 250W of power rating while the 4400 provides up to 437W. ISR4461 also supports dual 650W DC reverse-air power supplies which are included in the NEBS installation kit.

It is important to note that when DC power supplies are installed on the router, PoE based modules may not be used.

More details about the DC power supplies and its capabilities can be found at the below url https://www.cisco.com/c/en/us/td/docs/routers/access/4400/hardware/installation/guide4400-4300/C4400_isr/FRUs_Modules.html#49534.

Product Performance and Scalability

The Cisco 4000 Family is built on a multicore CPU architecture. It runs modular Cisco IOS XE Software, which allows the platform to use to full advantage a distributed multicore architecture. The architecture of the Cisco 4000 separates control- and data-plane operations and integrates an industry-first services plane. This design delivers full-featured integrated services up to Layer 7 at high performance with the ability to deliver application-aware network services while maintaining a stable platform and a high level of performance during periods of heavy network traffic.

The ISR 4000 consists of 3 series of Routers – the ISR 4400 series, the 4300 series and ISR 4200 series whose performance levels are represented by the chart below.

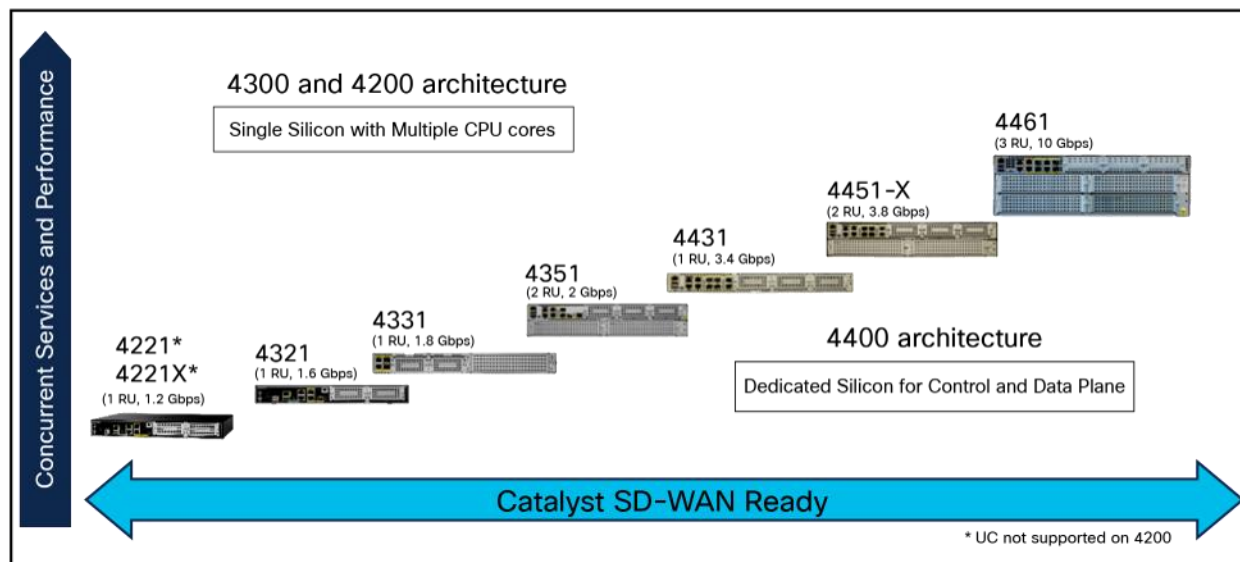


Figure 2.
ISR 4000 performance levels by series

ISR 4000 Interfaces and Modules Support

The Cisco 4000 Series Integrated Services Routers (ISRs) are modular routers with LAN and WAN connectivity. The routers provide for Network Interface Module (NIM) slots and enhanced Service Module (SM-X) slots offering a rich set of modules, such as LAN, WAN and wireless interfaces plus a range of compute engines for embedded services.

For the complete list of supported modules on the ISR4000 Series refer to the Interfaces and Modules sheet at <https://www.cisco.com/c/en/us/products/routers/4000-series-integrated-services-routers-isr/relevant-interfaces-and-modules.html>.

Cisco Catalyst SD-WAN with the ISR4000 Series

The ISR 4000 series is optimized for Cisco Catalyst SD-WAN. For enterprises this means that business critical applications run faster, with more reliability and reduced Operational Expenditure (OpEx). Cisco Catalyst SD-WAN achieves this by giving all branches and data centers the ability to monitor, control, move and report on streams of application data, such as specific web (HTTP) traffic for example. The ISR 4000 series has deep packet inspection capability and can accurately identify and control thousands of different applications including custom in-house enterprise applications.

The benefits are immense:

1. Business-critical applications no longer have to contend each other or with traffic that should be served on best effort basis.
2. The Enterprise network becomes more reliable because multiple paths can be used.
3. Costs are greatly reduced because dual MPLS links can be replaced with a mix of MPLS and Internet.

-
4. The time to bring up new remote sites is dramatically reduced because Catalyst SD-WAN supports rapidly deployed DSL and 3G/4G LTE connections as easily as MPLS.
 5. Security is assured across these connections using a zero-touch secure VPN technology used by governments and finance organizations worldwide.

From a platform perspective, the ISR 4000 series has

1. Separated control and data planes for Denial of Service (DoS) attack prevention and Intrusion Prevention System (IPS) and firewall capability built-in.
2. SaaS applications can have content locally cached. The caching is automatic and peers directly with Akamai technology to obtain intelligence.
3. Application performance speed is greatly increased using in-built application acceleration technology that can locally cache at a byte-level.

Cisco Catalyst SD-WAN implementation on the ISR4000 is managed at the end device either from the Cloud, or On-Premise, through ascending levels of Cisco DNA throughput-based subscription licenses. Cisco DNA Essentials and Advantage subscription licenses enable all customers to seamlessly transition between On-Premise and Cloud management as needed. The Cisco DNA subscription license tiers are structured to support the growth in business needs and simplify the journey to intent-based networking for the WAN.

For more information on Cisco DNA Subscriptions for SD-WAN and Routing, please refer to <https://www.cisco.com/site/us/en/products/networking/wan-software/dna-subscription-wan/index.html>.

Cisco Security Solutions for the ISR4000 Series

Cisco WAN MACsec

Cisco routers support a wide-range of ever enhancing security features on the ISR4000 routers. Cisco WAN MACsec is supported on the ISR4000 series routers using the NIM 2GE-CU-SFP module. WAN MACsec provides a line-rate network encryption solution over Layer 2 Ethernet transport services and can be leveraged outside campus networks, whether it be over Metro Ethernet transport or Data Center Interconnect (DCI) links. MACsec also secures WAN connections that are leveraging Ethernet as the link-layer media.

Cisco Secure Network Analytics

Cisco Secure Network Analytics provides enterprise-wide network visibility to detect and respond to threats in real-time. The solution continuously analyzes network activities to create a baseline of normal network behavior. It then uses this baseline, along with non-signature-based advanced analytics that include behavioral modeling and machine learning algorithms, as well as global threat intelligence to identify anomalies and detect and respond to threats in real-time. Secure Network Analytics can quickly and with high confidence detect threats such as Command-and-Control (C&C) attacks, ransomware, Distributed-Denial-of-Service (DDoS) attacks, illicit cryptomining, unknown malware, and insider threats. With an agentless solution, you get comprehensive threat monitoring across the entire network traffic, even if it's encrypted. For more information and platform support refer to <https://www.cisco.com/c/en/us/products/collateral/security/stealthwatch/datasheet-c78-739398.html>.

Cisco Snort IPS and Cisco Umbrella Branch

Cisco® Snort® IPS for Cisco 4000 Series Integrated Services Routers (ISRs) offers a lightweight threat defense solution that uses industry-recognized Snort open-source Intrusion Prevention System (IPS) technology. It is perfect for customers who are looking for a cost-effective solution that provides one box for both advanced routing capabilities and integrated threat defense security to help comply with regulatory requirements.

Cisco Umbrella Branch is a cloud-delivered security service for the Cisco Integrated Services Router (ISR). It provides the first layer of defense against threats at branch offices. And it offers the simplest, fastest way to protect every device on your branch network. You gain visibility and enforcement at the DNS layer, so you can block requests to malicious domains and IPs before a connection is ever made.

When enabling MACsec, you will need to procure the Security and HSEC licenses to stay within the limits of federal export control regulations. When customers wish to enable Cisco Secure Network Analytics, the Security (SEC) license needs to be enabled. Enabling Snort needs a Security (SEC) license and a signature subscription license. Enabling Cisco Umbrella Branch needs an Umbrella Branch license and a subscription license.

Reducing Operational Costs using Cisco ISR

Support for Data Modelling

Enterprises and Service Providers wish to drive down the operational cost (opex) of their networks and increase the agility and speed with which they deliver new services furthered by investments in Software Defined Networking (SDN) and Network Function Virtualization (NFV). Cisco routers provide support for Netconf and YANG data-modelling with increasing model coverage in successive releases.

Software Maintenance Upgrades (SMU)

The ISR4000 routers now support Software Maintenance Upgrades (SMU)⁷. An SMU is a package that can be installed on a system to provide a patch fix or security resolution to a released image, is provided on a per release and per component basis, and is specific to the platform. An SMU is an independent and self-sufficient package and does not have any prerequisites or dependencies. Please refer to <https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/config-mgmt/configuration/xe-16-6/config-mgmt-xe-16-6-book/cm-smu.html> for additional details.

Network Plug and Play

Network Plug and Play helps automate the onboarding of new devices on your network by applying configuration settings without manual intervention. With the ease of a centrally managed controller, it reduces the time a new device takes to join your network and become functional. For more information on Plug and Play (PnP), please refer to <https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Plug-and-Play/release/notes/pnp-connect-release-notes.html>.

Cisco IOS Software Licensing and Packaging

Universal IOS XE and XE-SDWAN Image

A single Cisco IOS XE Universal image encompassing all functions is delivered with the platform. Advanced features can be enabled by simply activating a software license on the Universal image. Technology

⁷ Supported from IOS-XE16.6.1 for all ISR4000 and ASR1000 series routers

packages and feature licenses, enabled through right-to-use licenses, simplify software delivery and decrease the operational costs of deploying new features.

Starting with Cisco IOS XE Release 17.2.1, you can install and deploy both Routing (Autonomous Mode) and Catalyst SD-WAN (Controller Mode) functionality on ISR4000 platforms. This feature supports the use of a single universalk9 image to deploy Cisco IOS XE in dual persona modes on all supported devices. This universalk9 image supports booting up in two modes - Autonomous mode (for regular IOS XE Routing features) and controlled mode (for Catalyst SD-WAN features).

When ordering an ISR router with a Cisco DNA Subscription, a customer may choose either Autonomous Mode (AUTO) for regular routing and Controller mode (CTRL) for Catalyst SD-WAN deployments. A license in Autonomous Mode is perpetual and doesn't require renewal, whereas a license in Controller Mode is term based and does require renewal upon expiry of the purchased license term. Refer to the Cisco DNA Ordering Guide at <https://www.cisco.com/c/en/us/products/collateral/software/dna-subscription-routing/guide-c07-740642.html>.

Also present is the High Security (HSEC) license, which removes the curtailment enforced by the U.S. government export restrictions on the encrypted tunnel count and encrypted throughput. The HSEC license is a separately required license for a feature to have full crypto functionality. Without the HSEC license, only 1000 secure tunnels and 250⁸ Mbps of crypto bandwidth would be available.

For additional information and details about Cisco IOS Software licensing and packaging on the Cisco 4000 Family, please visit

<https://www.cisco.com/c/en/us/td/docs/routers/access/4400/software/configuration/guide/isr4400swcfg.pdf>.

IOS-XE provides support for both perpetual and subscription licensing. Subscription Licensing with support for Cisco Catalyst Center is offered using Cisco DNA Essentials or Cisco DNA Advantage. Please refer to the Cisco DNA Ordering Guide at: <https://www.cisco.com/c/en/us/products/collateral/software/dna-subscription-routing/guide-c07-740642.html>.

Software Defined Networks may be provisioned through the two major licenses on the Cisco 4000 Family; these licenses can be activated through the Cisco software activation process identified at <https://www.cisco.com/go/sa> using the Cisco Catalyst Center or through the Cisco Catalyst SD-WAN Manager management portal. The XE-SDWAN image provides for its own licensing schema through Cisco DNA Licensing.

For more information on Cisco DNA Subscriptions for SD-WAN and Routing, please refer to <https://www.cisco.com/site/us/en/products/networking/wan-software/dna-subscription-wan/index.html>.

⁸ The change to 250Mbps was achieved in the IOS-XE version 16.8.1 pursuant to revised Federal regulations

Smart Software Licensing Support for IOS-XE

IOS-XE supports Smart Licensing beginning with image version 16.6.1 and Device Led Conversion with image version 16.9.1. Cisco Smart Licensing is a flexible licensing model that provides you with an easier, faster, and more consistent way to purchase and manage software across the Cisco portfolio and across your organization. And it is secure – you control what users can access. With Smart Licensing you get:

- **Easy Activation:** Smart Licensing establishes a pool of software licenses that can be used across the entire organization—no more PAKs (Product Activation Keys).
- **Unified Management:** Cisco License Central (CLC) provides a complete view into all of your Cisco products and services in an easy-to-use portal, so you always know what you have and what you are using.
- **License Flexibility:** Your software is not node-locked to your hardware, so you can easily use and transfer licenses as needed.

To use Smart Licensing, you must first set up a Smart Account on [Cisco License Central](#).

One or more Virtual Accounts maybe created under the Smart Account to enable the organization to logically segregate the purchased licenses. Device Led Conversion (DLC) allows the customer to convert all existing PAK and RTU licenses on the router into a Smart License.

For a more detailed overview on Cisco Licensing, visit the [Licensing Hub](#).

Cisco ISR 4000 bundles

Cisco ISR 4000 is available is several attractive bundles. The AX bundles integrate Cisco Wide Area Application Services (WAAS), Security (SEC), and Data (DATA) licenses into a single bundle that is simple to order, configure, and deploy. For customers who are also interested in voice along with all of these features, AXV presents an attractive option. See Table 6 for details.

Table 6. Cisco ISR 4000 feature bundles

Bundles	Features
Application Experience with Voice (AXV)	AX + Voice
Voice with Security (VSEC)	Voice + Security
Application Experience (AX)	IP Base + Security + advanced networking protocols: L2TPv3, BFD, MPLS, VRF, VXLAN ⁹ (Bandwidth <100Mbps) Application Experience: PfRv3, WAAS with AppNav, NBAR2, AVC, IP SLA Hybrid Cloud Connectivity: LISP, OTV (for Bandwidth <100Mbps), VPLS, EoMPLS Intelligent Web Caching: Akamai Connect
Voice (V)	IP Base + Unified Communications: CME, SRST, CUBE

⁹ Supported only on the ISR4451

Bundles	Features
Security (SEC)	IP Base + Advanced Security: Zone Based Firewall, IPSec VPN, EZVPN, DMVPN, FlexVPN

Note: ISR4221/K9 does not support UC (Voice), hence no V, VSEC, AXV bundles for ISR4221/K9

More information on ISR AX bundles is available at <https://www.cisco.com/go/ax>.

A pay-as-you-grow licensing model lets you increase the performance level for the platforms from the base level to a higher level. So you can purchase at an attractive entry-level price point and increase the performance level as your business demand grows. Table 7 describes the performance licenses.

Table 7. Cisco ISR 4000 performance licenses

Platform	Performance-on-Demand License	Features
ISR4461	FL-4460-PERF-K9	Increases the performance from base performance 1.5 Gbps to 3 Gbps
ISR4451	FL-44-PERF-K9	Increases the performance from base performance 1 Gbps to 2 Gbps
ISR4431	FL-44-PERF-K9	Increases the performance from base performance 500 Mbps to 1 Gbps
ISR4351	FL-4350-PERF-K9	Increases the performance from base performance 200 Mbps to 400 Mbps
ISR4331	FL-4330-PERF-K9	Increases the performance from base performance 100 Mbps to 300 Mbps
ISR4321	FL-4320-PERF-K9	Increases the performance from base performance 50 Mbps to 100 Mbps
ISR4221	FL-4220-PERF-K9	Increases the performance from base performance 35 Mbps to 75 Mbps

Table 8. Cisco ISR 4000 Booster (boost) performance licenses,

Part number	Description
FL-4220-BOOST-K9 (=)	Booster Performance License for 4220 Series Router for up to 1.2 Gbps CEF*
FL-4320-BOOST-K9 (=)	Booster Performance License for 4320 Series Router for up to 1.6 Gbps CEF*
FL-4330-BOOST-K9 (=)	Booster Performance License for 4330 Series Router for up to 1.8 Gbps CEF*
FL-4350-BOOST-K9 (=)	Booster Performance License for 4350 Series Router for up to 2 Gbps CEF*
FL-4430-BOOST-K9 (=)	Booster Performance License for 4430 Series Router for up to 3.4 Gbps CEF*
FL-44-BOOST-K9 (=)	Booster Performance License for 4450 Series Router for up to 3.8 Gbps CEF*
FL-4460-BOOST-K9 (=)	Booster Performance License for 4460 Series Router for up to 10 Gbps CEF*

* Test results for IP Routing (CEF) @ IMIX

Ordering information

The Cisco ISR 4000 Family is orderable and shipping. To place an order, refer to Table 9 below and visit the [Cisco Ordering Home Page](#).

Table 9. Cisco ISR 4000 Series ordering information

Product Name	Product Description
ISR4461/K9	Cisco ISR 4461 with 4 onboard GE, 3 NIM slots, 1 ISC slot, 3 SM slots, 8 GB Flash Memory default, 2 GB DRAM default (data plane), 4 GB DRAM default (control plane)
ISR4451-X/K9	ISR 4451 with 4 onboard GE, 3 NIM slots, 1 ISC slot, 2 SM slots, 8 GB Flash Memory default, 2 GB DRAM default (data plane), 4 GB DRAM default (control plane)
ISR4431/K9	ISR 4431 with 4 onboard GE, 3 NIM slots, 1 ISC slot, 8GB Flash Memory default, 2 GB DRAM default (data plane), 4 GB DRAM default (control plane)
ISR4351/K9	ISR 4351 with 3 onboard GE, 3 NIM slots, 1 ISC slot, 2 SM slots, 4 GB Flash Memory default, 4 GB DRAM default
ISR4331/K9	ISR 4331 with 3 onboard GE, 2 NIM slots, 1 ISC slot, 1 SM slots, 4 GB Flash Memory default, 4 GB DRAM default
ISR4321/K9	ISR 4321 with 2 onboard GE, 2 NIM slots, 1 ISC slot, 4 GB Flash Memory default, 4 GB DRAM default
ISR4221/K9	ISR 4221 with 2 onboard GE, 2 NIM slots, 1 ISC slot, 8 GB Flash Memory default, 4 GB DRAM default
ISR4221X/K9	ISR 4221 with 2 onboard GE, 2 NIM slots, 1 ISC slot, 8 GB Flash Memory default, 8 GB DRAM default

For additional product numbers, including the Cisco 4000 Family bundle offerings, please contact your local Cisco account representative. To place an order, visit the [Cisco Ordering Home Page](#). To download software, visit the [Cisco Software Center](#).

Integrated Services Router Migration Options

The Cisco ISR 4000 Family is included in the standard Cisco Technology Migration Program (TMP). Refer to <https://www.cisco.com/go/tmp> and contact your local Cisco account representative for program details.

Warranty information

The Cisco ISR 4000 Series Integrated Services Routers have a 90-day limited liability warranty.

Product sustainability

Information about Cisco's Environmental, Social and Governance (ESG) initiatives and performance is provided in Cisco's CSR and sustainability [reporting](#).

Table 10. Product sustainability

Sustainability Topic		Reference
General	Information on product-material-content laws and regulations	Materials
	Information on electronic waste laws and regulations, including our products, batteries and packaging	WEEE Compliance
	Information on product takeback and resuse program	Cisco Takeback and Reuse Program
	Sustainability Inquiries	Contact: csr_inquiries@cisco.com
	Environmental specifications	Table 5. Product specifications
	Regulatory and compliance	Table 5. Product specifications
	MTBF	Table 5. Product specifications
Power	Power supplies	Table 5. Product specifications
	Power	Table 5. Product specifications
Material	Product packaging weight and materials	Contact: environment@cisco.com
	Size and Weights	Table 5. Product specifications
	Packaging Weight	Table 5. Product specifications

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Document history

New or revised topic	Described In	Date
Completely revised and updated to newest standard	- Updated product family to include Cisco 4331-DC and Cisco 4221(X) - Updated General features (Table 1) - Updated Architectural highlights (Table 2) - Updated Specifications (Table 5) - Updated Cisco DNA Software for SD-WAN and Routing subscription tiers and licensing information - Updated Product Performance and Scalability (Figure 2) - Updated Boost performance license description to include throughput (Table 8) - Updated product names to reflect current state, i.e., Cisco Catalyst Center, Cisco Catalyst SD-WAN Manager, Cisco Catalyst SD-WAN, Cisco Secure Network Analytics, etc.	August 28, 2024
Completely updated to newest standard	Entire document	August 20, 2024
Revised content on Smart Licensing	Smart Software Licensing Support for IOS-XE	May 25, 2021
Change 4461 from 2RU to 3RU	Sizing Section	December 4, 2018
Added 4461 and SD-WAN		November 13, 2018
Added support for Cisco SD-WAN with IOS XE SD-WAN, 16.9.1 software release. Added Encrypted Traffic Analytics as well as Boost Performance Licensing support information.	Ordering Information	August 8, 2018
Added new ISR 4000 Series model, the ISR 4221. Updated all related modules, memory, licenses and bundles support related to the ISR 4221.	Features and Benefits	August 23, 2017

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